TEXORA · LECTURE SHEET

Machine Erection

A precision textile machine bolted to a bad foundation will vibrate, wear and produce faulty material for its whole life — no setting can cure it. Erection is the one-time operation that decides decades of performance. This sheet covers floor preparation, foundations, fixation and levelling as one sequence.

TE-4105 · INDUSTRIAL UTILITIES · TOPIC 8 · LEVEL 4 TERM 1

1. What erection includes

Machine erection (installation) is the complete process of receiving a machine, positioning it at its planned location, mounting it on its foundation, levelling and aligning it, connecting services (power, steam, water, air, drainage), and proving it by trial run. It is done from the maker's erection drawing and manual, which specify foundation size and depth, bolt positions, total and per-foot loads, levelling tolerances and service connections. Layout position comes from the mill plan (Topic 10): material flow, gangways and future expansion are fixed at this stage.

2. Floor preparation

- **Check the floor's duty:** distinguish machines standing on a ground-bearing industrial floor from those needing independent foundation blocks (§3). Confirm floor thickness, reinforcement and load capacity against the machine's static load plus dynamic (vibration/unbalance) allowance.
- **Clean and survey:** remove laitance, oil and debris; check the floor for level and flatness over the machine footprint; mark the machine centre lines and reference (datum) lines on the floor from the layout drawing with chalk/paint, and transfer bolt positions from the template.
- **Prepare service trenches and pits** where the design requires them (cable trenches, drain channels for wet processing, pits for elevators or loom under-motion where applicable) before the machine arrives.
- **Access:** confirm the route from gate to position — door widths, turning space, floor loading on the route, lifting gear (Topic 5) — before delivery day.

3. Foundation

A **foundation** supports the machine, spreads its load safely into the ground, keeps it level, and absorbs or isolates vibration.

- **Functions:** bear static weight; resist dynamic forces without rocking or settlement; damp vibration; protect the machine from ground-borne vibration of neighbours (and them from it).
- **Common form:** a reinforced-concrete block, sized from soil bearing capacity and machine load, cast in a prepared excavation with anchor-bolt pockets or a bolt template held exactly to the drawing. Heavier or high-speed machines need deeper, more massive blocks; vibration-sensitive neighbours may justify isolation pads (cork, rubber, spring isolators) between block and soil or machine and block.
- **Materials and curing:** concrete of the specified grade, properly compacted and **cured** (kept moist, typically at least 7 days and longer per specification before full loading — follow the project specification). Loading green concrete is a classic cause of later settlement and cracked grout.
- **Soil:** made-up (filled) ground must be compacted or the foundation taken down to firm strata; water table and drainage decide pit design.

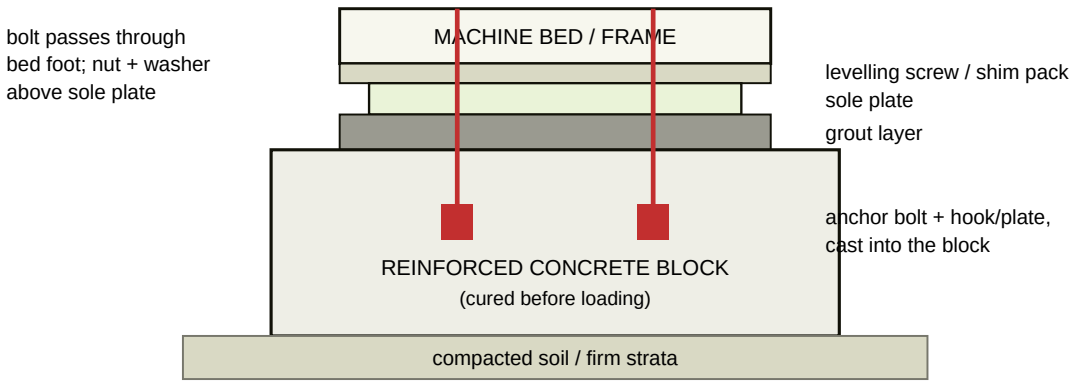


Figure 1 — Machine foundation cross-section. From top: machine bed, levelling screws/shims, sole plate, grout, reinforced concrete block, compacted soil. Anchor bolts (red) tie the bed to the block.

4. Machine fixation

Fixation fastens the machine to its foundation so it cannot shift, rock or walk under vibration:

1. Set the machine (or its sole plates) over the anchor bolts using lifting gear; lower onto levelling screws or shim packs, nuts finger-tight.
2. Level and align (§5) *before* final tightening.

3. **Grout:** pack or pour non-shrink grout under the sole plate/bed so the whole footprint bears evenly — voids under a foot concentrate load and crack.
4. After grout curing, tighten anchor nuts in a cross (diagonal) sequence, evenly, to the specified torque; re-check level after tightening — tightening itself can pull a machine out of level.
5. Fit dowels or chocks where the maker specifies positive location against lateral shift.

Light machines (sewing workstations, small auxiliaries) may be fixed with expansion bolts into a sound floor slab or stand on anti-vibration mounts without individual blocks — the principle (even bearing, no rocking, level) is unchanged.

5. Levelling

Levelling sets the machine's reference surfaces horizontal (and its shafts/rollers mutually aligned) within the maker's tolerance, using a precision spirit level (or laser/electronic level) on the machined datum surfaces named in the manual — never on painted or cast-rough surfaces.

Method, step by step: (1) Clean datum surfaces and the level; reverse the level end-for-end at each reading to cancel level error. (2) Level longitudinally, then transversely, adjusting levelling screws/shims at each foot a little at a time in sequence. (3) Check for a **soft foot**: with nuts slackened in turn, a foot that lifts or a dial that jumps shows uneven bearing — correct with shims, never by bolt force. (4) Align coupled shafts and connected machines (straightedge, dial indicator or laser) after individual levelling. (5) Re-check after grouting, after tightening, and again after the trial run and first days of operation, recording the readings in the machine file.

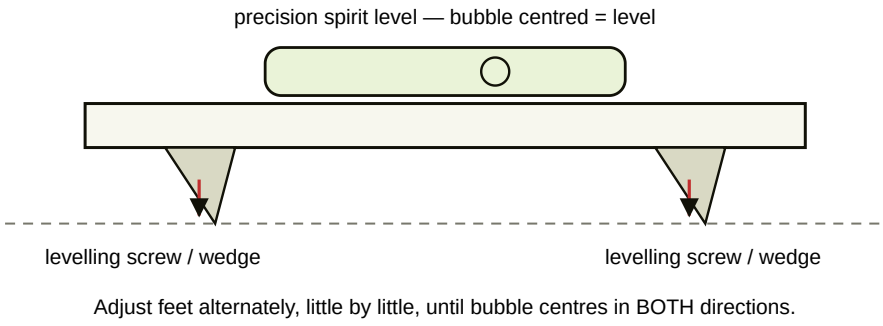


Figure 2 — Levelling sketch. The precision level sits on the cleaned datum surface; levelling screws/wedges at the feet are adjusted alternately until the bubble centres longitudinally and transversely.

6. Erection sequence — summary

1. **Plan:** layout drawing, maker's foundation drawing, services plan; access route checked.

2. **Prepare:** excavation/floor works, foundation cast with bolt template, cured; trenches and drains ready.
3. **Receive and position:** unpack, inspect for transit damage, move and set over bolts.
4. **Level and align:** rough level → grout → cure → final level, soft-foot check, shaft alignment.
5. **Fix:** tighten anchors in cross sequence; re-verify level.
6. **Connect:** power, steam/water/air, drainage, guards and safety devices fitted (Topic 11).
7. **Prove:** hand-turn, no-load run, load trial, product quality check; record readings; hand over with documents to production and maintenance (Topic 9).

Glossary

Foundation — the structure carrying a machine and spreading its load to the ground. **Anchor bolt** — bolt cast into the foundation to fasten the machine. **Grout** — flowable mortar packed under a machine base for full-area bearing. **Sole plate** — a steel plate under the machine foot distributing load to the grout. **Soft foot** — a machine foot not bearing evenly, distorting the frame when tightened. **Datum surface** — the machined reference surface specified for levelling. **Curing** — keeping concrete moist while it gains strength.

Key points

• Erection quality is permanent: settlement, soft foot and misalignment cannot be adjusted away later.
• Concrete must be cured before loading; bolts are set by template from the maker's drawing.
• Grout gives full-footprint bearing; tighten anchors in cross sequence and re-check level afterwards.
• Level on datum surfaces only; correct soft foot with shims, never with bolt force.

Practice

1. List the erection sequence for a new dyeing machine from delivery to handover, in order.
2. A newly erected machine vibrates and its level has drifted after one month. Give three erection faults that could explain this.
3. Explain soft foot: how it is detected and why tightening the anchor nut is the wrong cure.
4. Why is grout used under the sole plate instead of setting the plate directly on the concrete?

Answer key

1. Position per layout → set on foundation bolts → rough level → grout → cure → final level/alignment (soft-foot check) → cross-sequence tightening + re-level → connect

steam/water/power/drains and guards → no-load then load trial with quality check → recorded handover.

2. Any three: foundation loaded before curing (settlement); grout voids under a foot; soft foot forced down by bolts (frame distortion relaxing over time); foundation undersized/on uncompacted fill; anchor nuts tightened unevenly without re-levelling.
3. Detection: slacken each foot nut in turn and watch the level/dial — a reading change shows that foot was sprung. Tightening forces the frame to twist to meet the floor, distorting alignment; shimming the gap restores bearing without distortion.
4. Cast concrete is never perfectly flat; grout flows to fill every irregularity, giving continuous contact and load spread that a rigid plate on rough concrete (point contact) cannot achieve.
